

Stage 07 — Outliers + Risk Assumptions

7.1 Why Outliers Matter (Especially in Finance)

Outliers can be **errors**, **rare events**, or **early warnings**. In finance, fat-tailed returns, liquidity dry-ups, and flash crashes mean that extreme values may be central to risk understanding rather than noise to delete. Treating them requires **assumptions** about what is “normal.”

7.2 Defining Outliers

- **IQR Rule:** Points outside $Q1 - 1.5 \cdot IQR$ or $Q3 + 1.5 \cdot IQR$. Robust to skew, quick to compute, widely used with boxplots.
- **Z-score:** Points with $|z| > \text{threshold}$ (often 3). Interpretable if data are roughly normal; can over-flag for heavy tails.
- **Winsorizing:** Replace extremes with chosen quantile values. Preserves row count; reduces leverage; but alters data.

Key Point: There is **no single correct** definition. Choose a method, state your **assumptions**, and test **sensitivity**.

7.3 Sensitivity Analysis

Ask: How much do my conclusions change if I change how I treat outliers?

- Compare summary stats with vs. without outliers.
- Refit a simple model after removal and after winsorizing.
- Report the differences (coefficients, R^2 , MAE) in a small table.

7.4 Risks & Misconceptions

- **Not all outliers are errors.** A crash day is rare but real.
- **Removing is not always better.** You might delete exactly the signal you seek.
- **Thresholds are choices.** Record them and justify with domain context.

7.5 Minimal Engineering Patterns

- Write **reusable functions** (detection, handling).
- Use **boolean masks** for clarity and testability.
- Keep a **sensitivity table** and save it (CSV).
- Document **assumptions** in code and README for reproducibility.

7.6 Worked Mini-Example (Conceptual)

1. Fit regression on all data → slope = 1.8.
2. Remove IQR outliers → slope increases to 2.0, MAE drops.
3. Winsorize → slope ~1.9, MAE moderate. Interpretation: Outliers were pulling the fit; removing reduces leverage, winsorizing softens extremes.

7.7 What to Write in Your Project Repo

- A `src/` module with your outlier functions and docstrings.
- A notebook that compares **all vs. filtered vs. winsorized**.
- A README section: the thresholds you chose, *why* you chose them, and the observed impact.

Mantra: *Assumptions → Method → Sensitivity → Documentation.*